

Design Doc: AI Competitive Intelligence Platform — V1

Status: Ready to build

Scope: Portfolio MVP for APM and new graduate PM applications

1. Summary

A user enters the URL of a software product. The platform researches the product, discovers and verifies up to five competitors, and generates a concise competitive report covering positioning, features, and pricing. Every factual claim links to a retrievable public source and displays when that source was accessed.

The product is built around one principle: an AI-generated competitive report is only useful when a PM can quickly verify it. V1 therefore optimizes for evidence quality, source traceability, and measurable accuracy instead of report length or technical complexity.

2. Problem

Product managers, founders, and job candidates repeatedly conduct the same manual research:

- Understand a product's audience, positioning, features, and pricing
- Identify its most relevant competitors
- Search competitor websites and pricing pages
- Normalize inconsistent product language into a comparison
- Preserve the sources behind important conclusions

A strong first pass can take hours and quickly becomes outdated. Generic AI tools reduce writing time, but they may rely on stale knowledge, identify irrelevant competitors, invent pricing, or produce citations that do not support their claims.

The core user problem is not generating more analysis. It is producing analysis that is fast to verify and credible enough to use in a product discussion.

3. Target user and job to be done

Primary user

An early-career product manager preparing a competitive readout for a roadmap, strategy, or interview discussion.

Job to be done

When I need to understand a product’s competitive position, help me create a reliable first-pass analysis quickly so I can spend my time evaluating implications instead of collecting and organizing evidence.

User needs

- Find relevant competitors, including companies the user may not already know
- Compare products across consistent dimensions
- Verify important claims without repeating the research
- Distinguish sourced facts from AI interpretation
- See when pricing and product information were retrieved

4. V1 product hypothesis

If the platform combines multi-strategy competitor discovery with claim-level source attribution, users will be able to create a defensible competitive analysis faster than through manual research while maintaining enough trust to use the output in a product discussion.

V1 tests three questions:

1. Can the system identify relevant competitors from only a product URL?
2. Can it extract and normalize features and pricing from public primary sources?
3. Can it generate a report in which at least 90% of factual claims are supported by working, relevant citations?

5. Goals and success measures

Goal	V1 success measure
Discover relevant competitors	At least 70% precision across a 10-product benchmark
Produce traceable analysis	At least 90% of factual claims have a working citation
Keep claims grounded	No more than 5% of factual claims are contradicted by their cited source
Extract usable pricing	At least 80% accuracy for publicly listed tiers and prices

Reduce research time

Complete a report in under 10 minutes

Control cost

Average model and search cost at or below \$1.20 per report

These are initial targets. Final résumé metrics must come from measured V1 results rather than estimates.

6. Non-goals

V1 will not include:

- Continuous competitor monitoring or scheduled recrawls
- Email, Slack, or other notifications
- Review sentiment analysis
- News, funding, headcount, or market-trend analysis
- Strategic recommendations presented as facts
- Team accounts, collaboration, authentication, or billing
- User-edited competitor lists
- A hosted vector database
- A distributed background-job system
- Support for every industry or complex enterprise product

V1 will focus on English-language B2B and consumer software products with publicly accessible websites and pricing information.

7. User experience

Primary flow

1. The user enters a product URL.
2. The platform extracts a structured product profile.
3. It displays the inferred category, audience, and value proposition.
4. It generates competitor candidates using multiple discovery strategies.
5. It verifies, deduplicates, ranks, and selects up to five competitors.
6. It researches primary product, feature, and pricing pages.
7. It generates the report and displays research progress.
8. The user opens citations to verify individual claims.

Report sections

V1 produces four sections:

1. **Executive summary:** A concise overview of the product and competitive landscape

2. **Competitive landscape:** Verified competitors, their positioning, and why they are relevant
3. **Feature comparison:** A normalized comparison of important capabilities
4. **Pricing comparison:** Public tiers, prices, billing periods, and limitations

Trust model

The interface distinguishes three content types:

- **Verified:** Extracted from a company's primary website or pricing page
- **Reported:** Supported by another publicly accessible source when a primary source is unavailable
- **Interpretation:** AI synthesis based on cited evidence and visually labeled as analysis

Every factual claim includes its source link and retrieval date. Missing or ambiguous information is displayed as unavailable instead of inferred.

8. Functional requirements

P0 requirements

- Accept and validate a public product URL
- Extract product name, category, audience, value proposition, features, and pricing
- Discover competitor candidates through at least three strategies
- Verify that each selected competitor has a reachable website and serves the inferred category or customer need
- Deduplicate candidates by normalized company name and domain
- Rank and return no more than five competitors
- Crawl selected public product, feature, and pricing pages
- Store raw source text, URL, retrieval time, and source type
- Extract structured findings from collected evidence
- Generate four report sections using structured output
- Attach source identifiers to factual claims
- Visually distinguish factual claims from interpretation
- Export or share the completed report through a stable link or PDF
- Record latency, estimated cost, tool calls, and failure states for each run

P1 requirements

- Show progress while research is running
- Allow the user to retry a failed page or report run
- Cache recently fetched pages to reduce cost and latency
- Show an explanation for why each competitor was selected

9. Competitor discovery

Competitor discovery is the primary product risk and the most important part of V1 to evaluate.

Discovery strategies

1. **Model-generated candidates:** Ask the model for likely competitors based on the structured product profile. These candidates are leads, not evidence.
2. **Search-generated candidates:** Search for phrases such as “alternatives to [product],” “[product] vs,” and “best [category] software.”
3. **Company comparison pages:** Search the product’s own website for pages comparing itself with other companies.

Verification and ranking

The system merges candidates by normalized name and domain. A candidate is retained only when its public website provides evidence that it addresses the same category, audience, or core customer problem.

Candidates are ranked using:

- Category overlap
- Target-customer overlap
- Feature overlap
- Frequency across discovery strategies
- Evidence strength

V1 selects no more than five competitors to keep research cost, report clarity, and evaluation manageable.

10. System workflow

Step 1: Product profiler

Fetch the submitted URL and a small set of discoverable pages such as pricing, features, about, and customers. Extract a structured profile containing:

- Product name and domain
- Product category
- Target customer
- Core customer problem
- Value proposition
- Key features
- Business model and pricing, when public

The user can see the inferred profile before the final report, making incorrect categorization easier to identify.

Step 2: Competitor discovery

Run the three discovery strategies, merge duplicate candidates, and create a ranked candidate set.

Step 3: Competitor verification

Visit each candidate's official website. Confirm that the company addresses the inferred category or customer problem. Remove candidates that cannot be verified.

Step 4: Evidence collection

For the original product and each verified competitor, collect text from a limited set of official pages:

- Homepage or primary product page
- Features page
- Pricing page

Each stored source includes its URL, retrieval time, HTTP status, raw text, and content hash.

Step 5: Structured extraction

Convert the collected evidence into normalized product findings. Each finding includes the source identifiers used to produce it.

Step 6: Report generation

Generate each report section from structured findings rather than from the model's general knowledge. The model must return structured claims containing:

- Claim text
- Claim type
- Supporting source identifiers
- Confidence

The interface renders these records into the final report.

Step 7: Validation

Before displaying the report, automatically flag:

- Factual claims without sources
- Citations whose source identifiers do not exist
- Pricing claims without a primary-source pricing page
- Recommendations or interpretations not labeled as analysis

11. Technical architecture

V1 stack

Layer	V1 choice	Reason
Frontend	Next.js, React, TypeScript	Fast development and a polished interactive demo
Backend	Python and FastAPI	Strong support for extraction, AI workflows, and evaluation scripts
Workflow	Explicit Python state machine or lightweight LangGraph graph	Makes steps observable and prevents uncontrolled agent loops
Database	SQLite during local development	Supports structured relationships with minimal setup
Source retrieval	Search API and public-page crawler	Provides current public evidence
AI	Larger model for profiling and synthesis, smaller model for extraction and classification	Balances output quality and cost
Deployment	Hosted frontend and backend with a managed relational database if required	Makes the project accessible to recruiters and interviewers

Why V1 does not require PostgreSQL or vector search

SQLite is sufficient for local development and the 10-product evaluation because V1 has one user, low data volume, and no concurrent workloads. The schema preserves a straightforward migration path to PostgreSQL when the application is deployed or supports multiple users.

Vector retrieval is intentionally excluded from the initial implementation. V1 collects a limited number of pages per product and converts them into structured findings before synthesis. This makes the pipeline easier to debug and the relationship between evidence and claims easier to evaluate. Vector search should be added only if testing shows that source volume exceeds model context limits or section-specific retrieval materially improves quality, latency, or cost.

Tool interface

V1 exposes four independently testable research tools:

Tool	Purpose
<code>search_web</code>	Return ranked public search results with URLs and snippets
<code>crawl_page</code>	Return cleaned text and retrieval metadata for a URL
<code>extract_pricing</code>	Return structured pricing tiers and source references
<code>find_alternatives</code>	Generate competitor candidates from search and comparison evidence

These tools may be exposed through MCP to demonstrate a portable tool interface, but MCP is not required for the core product to function.

12. Data model

products

`id, url, domain, name, category, profile_json, created_at`

runs

`id, product_id, status, started_at, finished_at, cost_cents, token_count, tool_calls, error`

competitors

`id, run_id, name, domain, relationship, confidence, discovery_methods_json, verified`

sources

`id, run_id, competitor_id, url, source_type, fetched_at, raw_text, http_status, content_hash`

findings

`id, run_id, competitor_id, dimension, value_json, source_ids_json, extracted_at`

claims

`id, run_id, section, text, claim_type, source_ids_json, confidence`

eval_results

`id, run_id, metric, score, details_json, created_at`

Claims are stored separately from rendered report text so citation coverage can be measured directly instead of estimated from prose.

13. Evaluation plan

The evaluation system is a core product feature, not an optional engineering task.

Benchmark set

Create a fixed set of 10 software products containing:

- Four established products with well-known competitors
- Three niche B2B products
- Two recently launched products
- One product with an ambiguous category

For every benchmark product, manually label:

- Acceptable product category
- Clearly relevant competitors
- Clearly irrelevant competitors
- Public pricing for the top three competitors on the evaluation date

Metrics

Metric	Definition
Competitor precision	Relevant or defensible competitors divided by all competitors surfaced
Category accuracy	Profiles matching the accepted category or synonym
Pricing accuracy	Correct public tier and price extractions divided by evaluated tiers

Citation coverage	Factual claims containing at least one retrievable citation
Citation validity	Sampled citations that directly support their associated claims
Hallucination rate	Sampled claims contradicted by their cited evidence
Latency	Time from URL submission to completed report
Cost per report	Search and model cost for a completed run

Evaluation approach

Deterministic checks run across all 10 benchmark products. Citation validity and hallucination rate are manually reviewed for a fixed sample of claims. V1 should publish the complete results table, including regressions and failures.

14. Key product decisions and tradeoffs

Optimize for trust over report breadth

V1 produces four well-supported sections instead of a broad strategy report. This tests whether users can trust the research before expanding the product's analytical depth.

Prefer structured extraction over vector retrieval

The product collects a bounded amount of evidence and converts it into normalized findings. This reduces architectural complexity and makes failures attributable to a particular extraction or synthesis step.

Limit the report to five competitors

A smaller set improves research depth, interface clarity, latency, cost, and evaluation quality. Users need a credible first-pass landscape, not an exhaustive market database.

Use primary sources for pricing

Pricing is extracted only from official company pages. If a price cannot be found or parsed, the platform reports it as unavailable rather than estimating it.

Keep interpretation visibly separate

AI synthesis can be useful without being treated as fact. Interpretations are labeled and linked to the evidence claims on which they are based.

15. Risks and mitigations

Risk	Mitigation
Irrelevant competitors	Use multiple discovery strategies, website verification, and a relevance-ranking rubric
Hallucinated claims or pricing	Generate from stored findings, require citations, and mark missing information unavailable
Dynamic or inaccessible websites	Fail softly, display unavailable data, and preserve source retrieval metadata
Excessive agent cost or looping	Use an explicit bounded workflow with tool-call and cost limits
Ambiguous competitor ground truth	Label candidates as clearly relevant, defensible, or clearly irrelevant and report the rubric
Scope expansion	Keep sentiment, news, monitoring, collaboration, and vector infrastructure outside V1

16. Build plan

Part 1: Core research pipeline

- Define schemas and SQLite data model
- Build URL validation and product profiling
- Implement search and crawl tools
- Implement competitor discovery, deduplication, verification, and ranking
- Test the pipeline on three products

Part 2: Grounded report experience

- Implement feature and pricing extraction
- Store sources, findings, and claims separately
- Generate the four report sections
- Add claim-level citations and trust labels
- Build the report interface and progress states

Part 3: Evaluation and portfolio polish

- Create and label the 10-product benchmark
- Measure precision, pricing accuracy, citation coverage, validity, latency, and cost
- Fix the highest-impact failure mode

- Deploy the application
- Publish the results table, failure gallery, README, and three-minute demo video

If time runs short, reduce visual polish or secondary report content before removing the evaluation suite.

17. Launch criteria

V1 is complete when:

- A user can submit a public software-product URL and receive a four-section report
- The system returns no more than five verified competitors
- Factual claims include retrievable citations and retrieval dates
- Pricing is sourced from official pages or marked unavailable
- The evaluation suite has run on all 10 benchmark products
- Actual quality, latency, and cost results are published
- The project has a live demo, public repository, concise writeup, and demo video

18. Portfolio and résumé deliverables

The finished portfolio should include:

1. A live product that works without local setup
2. This design document, updated with decisions that changed during development
3. A GitHub repository with a clear architecture and evaluation README
4. An evaluation table showing actual performance across versions
5. A failure gallery explaining three meaningful failures and resulting product changes
6. A three-minute demo focused on the user problem, product decisions, and measured results

Résumé bullet template after V1 is complete

- Built and launched an AI competitive intelligence platform that transformed a product URL into an evidence-backed landscape, feature, and pricing analysis across up to five verified competitors in **[measured latency]**.
- Designed a multi-step research workflow with automated competitor discovery, structured web extraction, and claim-level citations, achieving **[precision]% competitor precision, [citation coverage]% citation coverage**, and **[pricing accuracy]% pricing accuracy** across a 10-product benchmark.
- Created an evaluation system for groundedness, hallucination rate, latency, and cost, then improved **[target metric]** from **[baseline]** to **[final result]** by **[specific product or technical change]**.

Do not place target or estimated numbers on the résumé. Replace every bracketed field only after running the benchmark and recording the actual result.

19. Future extensions

Only after V1 is complete and evaluated, potential extensions include:

- User confirmation or editing of the competitor set
- Review sentiment and customer-pain analysis
- News and positioning changes
- Strategic threats and recommendations
- Scheduled monitoring and change alerts
- PostgreSQL and vector retrieval for larger, multi-user workloads

These extensions are deliberately excluded from the initial résumé artifact because a narrow product with demonstrated quality is more credible than a broad product with unverified output.